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PSQT

1 B

There are 5 basic steps in hypothesis testing.

(i) set Null Hypothesis

(ii) set Alternative Hypothesis

(iii) set level of significance

(iv) test statistic

(v) draw conclusions.

C

Level of confidence + level of significance = 1

Level of confidence is the probability that the hypothesis won't fail and it is steady.

level of significance is the probability of possible risk for the hypothesis to be proven.

D1

Queue.

A queue is of First Come First Serve.

The person who has been waiting at the bank will eventually move forward to the counter and have his service completed.

The person added is called ~~pushed~~<sup>enqueue</sup> into the queue.  
The person out of queue is called ~~popped~~ dequeue.

E

Waiting line theory

The waiting line theory can be used to bring uniformity in places so that their system can work uniformly.

- Train Ticket Booking
- Barber Shop

F

Markov chains.

There are 4 models in Q.T.

1)  $M/M/1/\infty$

2)  $M/M/1/N$

3)  $M/M/s/\infty$

4)  $M/M/s/N$

For each in it the  $M$  is for

$M \rightarrow$  Markov poisson average

$M \rightarrow$  Markov exponential mean.

A1

P-value approach



2 A

- Types of errors

There are 2 types of errors.

TYPE I ERROR  $\rightarrow$

Null Hypothesis

$H_0$  is wrongly judged wrong even if it's true.

$H_0$  is a very true fact Hypothesis but instead of accepting it, they reject it accepting  $H_1$ .

TYPE II ERROR  $\rightarrow$

$H_0$  is accepted even though it is wrong.

$H_0$  is not true and should be rejected but instead, false Hypothesis is accepted.

B

$$H_0; \mu = 20\%$$

$$H_1; \mu \neq 20\%$$

Since  $n = 60 > 30$

Z test

$$\bar{x} = 25\%$$

$$s = 5\%$$

$$n = 60$$

$$Z = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$\frac{0.25 - 0.20}{0.05/\sqrt{60}} = \frac{0.05 \times \sqrt{60}}{0.05}$$

$$= \sqrt{60} = 7.745$$

$$\text{Level of significance} = 0.05$$

$$Z_{\alpha/2} = 1.96$$

C]

Queueing model.

M/M/1/FIFS.

M/M/1/∞

In this model there is only 1 server and ∞ number of <sup>in</sup> Queue.

The only 1 server has to attend the service for ∞ queue. never ending.

M/M/1/N

In this model there is only 1 server and has limited n number of people in Queue.

Only 1 server has to attend the service of N people waiting in Queue.

D]

N  
Server - 95 4 1  
Someton - 96 90 4  
Always - 0 10 90

$$\begin{bmatrix} 95 & 4 & 1 \\ 6 & 90 & 4 \\ 0 & 10 & 90 \end{bmatrix}$$

$$[x, y, z] \begin{bmatrix} 95 & 4 & 1 \\ 6 & 90 & 4 \\ 0 & 10 & 90 \end{bmatrix} = x, y, z$$

$$95x + 6y + 0z = x$$

$$x + y + z = 1$$

$$4x + 90y + 10z = y$$

$$1x + 4y + 90z = z$$



$$95x + 6y + 0z = x$$

$$4x + 90y + 10z = y$$

$$1x + 4y + 90z = z$$

$$x+y+z=1$$

$$94x = -6y$$

$$89y = -4x - 10z$$

$$x = \frac{-6y}{94}$$

$$89z = -4y - 1x$$

$$89z = -4y + \frac{6y}{94}$$

$$z(89) = \frac{-(94)4y + 6y}{(94)(89)}$$

$$89y = \frac{+4(6y)}{94} + 10 \frac{(-(94)4y + 6y)}{94(89)}$$

$$= \frac{12y}{94} + \frac{3820y}{94}$$

4 A

$$n = 50$$

A

B

$$\lambda = 86.7$$

$$\lambda = 77.8$$

$$s = 6.28$$

$$s = 5.67$$

Level of significance = 0.05

$$Z = \frac{86.7 - 12}{6.28/\sqrt{50}}$$

$$Z = \frac{77.8 - 12}{5.67/\sqrt{50}}$$

$$= \frac{74.7}{6.28} \times 7.07$$

$$= \frac{65.8}{5.67} \times 7.07$$

$$= 84.096$$

$$= 82.046$$

$$84.096 > 82.046$$

B

|    | OT  |     |     |
|----|-----|-----|-----|
|    | 23  | 60  | 29  |
|    | 28  | 79  | 60  |
|    | 9   | 49  | 63  |
| CT | 60  | 188 | 152 |
|    | 400 |     |     |

GT

 $\frac{\text{Pract}}{OT}$ 

| $O_i$ | $e_i$ | $O_i - e_i$ | $(O_i - e_i)^2$ |
|-------|-------|-------------|-----------------|
| 23    | 16.8  | 6.20        | 38.44           |
| 28    | 25.05 | 2.95        | 8.70            |
| 9     | 18.15 | -9.15       | 83.72           |
| 60    | 52.64 | 7.36        | 54.16           |
| 29    | 78.44 | 0.56        | 0.31            |
| 49    | 56.87 | -7.87       | 61.93           |
| 29    | 42.56 | -13.56      | 183.87          |
| 60    | 63.46 | 3.46        | 11.97           |
| 63    | 45.98 | 17.02       | 289.68          |

$$LOS = 0.01$$

21.67

A

$$\lambda = 12/\text{hr.}$$

$$\frac{1}{\mu} = 5$$

$$\mu = \frac{1}{5} \text{ min.} = \frac{1}{5} \times 60 = 30/\text{hr.}$$

$$\rho = \frac{\lambda}{\mu}$$

$$= \frac{12}{30} = \frac{2}{5}$$

$$L_s = \frac{\lambda}{\mu - \lambda} = \frac{12}{30 - 12} = \frac{12}{18} = 0.66$$

$$L_Q = \frac{\lambda^2}{\mu(\mu - \lambda)} = \frac{(12)^2}{30(18)} = \frac{144}{540} = 0.266$$



$$W_S = \frac{L_S}{\lambda} = \frac{1}{\mu - \lambda} = \frac{1}{30 - 12} = \boxed{\frac{1}{18}} = 0.055$$

$$W_q = \frac{L_q}{\lambda} = \frac{\lambda}{\mu(\mu - \lambda)} = \frac{12}{30(18)} = \boxed{\frac{12}{540}} = 0.022$$

B

$$P_2 = \begin{bmatrix} 0.9 & 0.03 & 0.07 \\ 0.15 & 0.80 & 0.05 \\ 0.20 & 0.30 & 0.50 \end{bmatrix}$$

$$P = 0.022$$

$$P_0 = \begin{bmatrix} 0.2 & 0.5 & 0.3 \end{bmatrix}$$

$$P_n = P_{n-1} [P]$$

$$P_2 = P_1 [P]$$

$$P_1 = P_0 [P]$$

$$P_1 = \begin{bmatrix} 0.2 & 0.5 & 0.3 \end{bmatrix} \begin{bmatrix} 0.9 & 0.03 & 0.07 \\ 0.15 & 0.80 & 0.05 \\ 0.20 & 0.30 & 0.50 \end{bmatrix}$$

$$= 0.05 + 0.075 + 0.5$$

$$\begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} 0.9 & 0.03 & 0.07 \\ 0.15 & 0.80 & 0.05 \\ 0.20 & 0.30 & 0.50 \end{bmatrix} = \begin{bmatrix} x & y & z \end{bmatrix}$$

$$0.9x + 0.15y + 0.20z = x$$

$$0.03x + 0.80y + 0.30z = y$$

$$0.07x + 0.05y + 0.50z = z$$





